Adaptations to the Physical Environment: Water and Nutrients

Chapter Concepts

- Water has many properties favorable to life
- Many inorganic nutrients are dissolved in water
- Plants obtain water and nutrients from the soil by the osmotic potential of their root cells
- Forces generated by transpiration help to move water from roots to leaves
- Salt balance and water balance go hand in hand
- Animals must excrete excess nitrogen without losing too much water

Water and Life

Physical Limits and Adaptations

- Organisms are restricted by their physical environments (e.g., oxygen and temperature for deep-diving mammals).
- Adaptations such as insulation (fat in seals, air in penguins) help maintain body temperature.
- Life depends on water as the basic medium and energy to power processes.
- Organisms affect and are affected by the physical world (e.g., oxygen in the atmosphere from photosynthesis).

Properties of Water Favorable to Life

- Water is liquid over most of Earth's surface temperatures.
- Water dissolves inorganic compounds, making it an excellent medium for chemical processes.
- Water's density and viscosity support organisms but also limit movement.
- Streamlined shapes and flotation adaptations (swim bladders, oil droplets, appendages) counteract sinking and drag.

Thermal Properties of Water

- Water resists temperature changes and remains liquid over a broad range.
- High energy required for evaporation and freezing stabilizes aquatic environments.
- Water expands and becomes less dense as it freezes, so ice floats.

Density and Viscosity

- Water is 800 times denser than air, providing buoyancy.
- High viscosity impedes movement, leading to adaptations in aquatic organisms.

Inorganic Nutrients in Water

Major Nutrients Required

- Organisms require hydrogen, carbon, oxygen, nitrogen, phosphorus, sulfur, potassium, calcium, magnesium, iron, sodium.
- Diatoms use silicates for shells; tunicates accumulate vanadium; nitrogen-fixing bacteria require molybdenum.

Solvent Capacity of Water

- Water dissolves minerals from rocks and soils, making nutrients available.
- Table salt ($NaCl$) dissolves into sodium (Na^+) and chloride (Cl^-) ions.
- Oceans concentrate ions over time; calcium carbonate precipitates as limestone.

Hydrogen Ions and pH

- Hydrogen ions (H^+) are highly reactive and affect enzyme activity.
- Acidity is measured as pH; pure water has pH 7.
- Most natural waters have pH between 6 and 9.

- Hydrogen ions help dissolve minerals, making nutrients available but can also mobilize toxic metals.

Water and Nutrient Uptake by Plants

Soil Structure and Water-Holding Capacity

- Soil particle size affects water retention; clay holds more water but less is available to plants.
- Water potential (matric potential) determines water availability; field capacity is the maximum available water.

Osmotic Potential and Water Uptake

- Water moves from high to low water potential.
- Osmosis allows plant roots to draw water from soil.
- Osmotic potential depends on solute concentration; desert and salt-adapted plants increase solute concentration to lower water potential.

Water Movement in Plants

Transpiration and Cohesion–Tension Theory

- Water moves from roots to leaves via xylem, driven by transpiration.
- Water potential gradient from leaves to roots pulls water upward.
- Stomates (stomata) control water loss and gas exchange.

Salt and Water Balance

Osmoregulation in Plants

- Plants pump excess salts back into the soil by active transport.
- Mangroves maintain high organic solute concentrations and have salt glands to excrete salt.

Osmoregulation in Animals

- Terrestrial animals retain ions and excrete excess salts in urine.
- Desert animals produce highly concentrated urine; birds and reptiles have salt glands.

Water Balance in Aquatic Animals

- Freshwater fish are hyperosmotic; they gain water and lose solutes, so kidneys and gills retain salts.
- Marine fish are hypo-osmotic; they lose water and must drink seawater, excreting excess salts.
- Sharks retain urea to match osmotic potential of seawater.

- Some marine invertebrates (e.g., Tigriopus) rapidly adjust osmotic potential by synthesizing or degrading amino acids.

Nitrogen Excretion

- Aquatic animals excrete ammonia (NH_3) rapidly in dilute urine.
- Terrestrial animals excrete urea ($CO(NH_2)_2$) or uric acid ($C_5H_4N_4O_3$) to conserve water.
- Excreting urea and uric acid is energetically costly.

Summary

- Water is the basic medium of life and a powerful solvent.
- Water resists temperature changes and provides buoyancy.
- All natural waters contain dissolved substances.
- pH affects nutrient availability and toxicity.
- Soil structure and water potential determine water availability to plants.
- Plants and animals have adaptations for water and salt balance.
- Nitrogen excretion strategies vary by environment.